

Disproof of Conjecture 00000001564: a k -dimensional cs-neighborly polytope can have only $2k < 2^{k+1}$ vertices

jilint777

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Abstract

Conjecture 00000001564 asserts that the minimal number of vertices of a k -dimensional cs-neighborly polytope is 2^{k+1} , and that the extremal bodies are the Hanner polytopes. The first clause is false for every $k \geq 1$. The k -dimensional cross-polytope $C_k^\Delta = \text{conv}\{\pm e_1, \dots, \pm e_k\}$ is centrally symmetric, has $2k$ vertices, and is cs-neighborly in the strongest possible sense: every set of its vertices containing no antipodal pair is the vertex set of a face. Since $2k < 2^{k+1}$, the claimed minimum is wrong; the smallest cases are the square ($4 < 8$) and the octahedron ($6 < 16$). The counterexample works under every reading of “cs-neighborly” we know, including the statement’s own definition. We also show that the true minimum is $2k$, and that no Hanner polytope has 2^{k+1} vertices. The refutation is formalized in Lean 4 (core library only) for $k = 1, 2, 3, 4$. The Chinese version of the statement asks about $2k$ -dimensional polytopes. That claim holds for $k = 1, 2$ and fails for every $k \geq 3$; the smallest counterexample is the 6-dimensional cross-polytope ($12 < 16$), also formalized in Lean.

1 The conjecture

Definition: A cs-neighborly polytope is centrally symmetric and has support points in every direction. *Conjecture:* The minimal number of vertices of a k -dimensional cs-neighborly polytope is 2^{k+1} ; and the extremal bodies are the Hanner polytopes.

The Chinese version differs in one point: it speaks of a “ $2k$ -dimensional” polytope (the dimension is written as “two k ”), while the number of vertices is still 2^{k+1} . We treat this reading separately in Section 5. The conjecture is a conjunction, so it is enough to refute the first clause:

$$\min\{f_0(P) : P \text{ a } k\text{-dimensional cs-neighborly polytope}\} = 2^{k+1} \quad (k \geq 1), \quad (1)$$

where $f_0(P)$ is the number of vertices of P . A statement “the minimum is n ” contains the lower bound “every such P has $f_0(P) \geq n$ ”. We refute this lower bound, which is the weakest part of (1).

Readings of “cs-neighborly”. The definition in the statement is unusual. “Having support points in every direction” holds for every nonempty compact set, so under the literal reading a cs-neighborly polytope is simply a centrally symmetric polytope. The standard notion is Grünbaum’s [1]. A centrally symmetric (cs) polytope P is *cs- j -neighborly* if every set of at most j vertices that contains no pair of antipodal vertices $\{v, -v\}$ is the vertex set of a face of P . “cs-neighborly” means cs-2-neighborly (any two non-antipodal vertices span an edge), or cs- $\lfloor k/2 \rfloor$ -neighborly for a k -polytope, depending on the author. We treat all these readings at once, via the strongest property:

(F) every nonempty set of vertices without an antipodal pair is the vertex set of a face.

Property (F) implies cs - j -neighborliness for every j , and it implies the statement's literal definition. The cross-polytope has property (F) (Theorem 1), so it is a counterexample under every reading.

2 Definitions

A *polytope* is the convex hull $P = \text{conv } V$ of a finite set $V \subset \mathbb{R}^k$. Its *dimension* is the dimension of its affine hull. A point $v \in V$ is a *vertex* of P if some linear functional c attains its maximum over P only at v . Every vertex of P lies in V . If V is in *convex position* (every point of V is a vertex), then $f_0(P) = |V|$. For a functional c , the set $F_c = \{x \in P : c \cdot x = \max_P c\}$ is a face of P . Its vertex set is $F_c \cap V = \text{argmax}_{v \in V} c \cdot v$, and every nonempty face of a polytope arises this way. So a set $S \subseteq V$ is the vertex set of a nonempty face iff $S = \text{argmax}_V c$ for some c . P is *centrally symmetric* if $P - x_0 = -(P - x_0)$ for some center x_0 . For $x_0 = 0$ this means $V = -V$ for the vertex set V .

3 The cross-polytope

Theorem 1. *Let $k \geq 1$ and $V_k = \{\pm e_1, \dots, \pm e_k\} \subset \mathbb{R}^k$, $C_k^\Delta = \text{conv } V_k$. Then:*

1. C_k^Δ is k -dimensional and centrally symmetric (about 0);
2. V_k is in convex position, so $f_0(C_k^\Delta) = 2k$;
3. every nonempty $S \subseteq V_k$ containing no antipodal pair is the vertex set of a face, namely $S = \text{argmax}_{V_k} c_S$ with $c_S = \sum_{s \in S} s$;
4. C_k^Δ has a support point in every direction.

Proof. (1) $V_k = -V_k$. The affine hull contains $0 = \frac{1}{2}(e_1 + (-e_1))$ and every e_i , so it is $\mathbb{R}^k$. (2) For $v \in V_k$ we have $v \cdot v = 1$ and $v \cdot w \in \{0, -1\}$ for $w \in V_k \setminus \{v\}$. So $c = v$ is maximized over V_k only at v , and hence over C_k^Δ only at v . (3) Since S contains at most one of $e_i, -e_i$, the i -th coordinate of c_S is $+1$ if $e_i \in S$, -1 if $-e_i \in S$, and 0 otherwise. Hence $c_S \cdot s = 1$ for $s \in S$. For $w \in V_k \setminus S$ we get $c_S \cdot w \in \{0, -1\}$: it is -1 if $-w \in S$ and 0 otherwise. So $\text{argmax}_{V_k} c_S = S$. (4) Any functional attains its maximum on the compact set C_k^Δ . $\square$

4 The disproof

Theorem 2. *For every $k \geq 1$ there is a k -dimensional centrally symmetric polytope with property (F) (hence cs - j -neighborly for every j , and cs -neighborly in the sense of the statement) that has $2k < 2^{k+1}$ vertices. Hence claim (1) is false for every $k \geq 1$, and so is the conjecture.*

Proof. Take C_k^Δ (Theorem 1). We have $2k \leq 2^k < 2^{k+1}$, since $2 \cdot 1 = 2^1$ and $2(k+1) = 2k + 2 \leq 2^k + 2^k$. $\square$

The smallest instances are the segment ($k = 1$: $2 < 4$), the square $\text{conv}\{(\pm 1, 0), (0, \pm 1)\}$ ($k = 2$: $4 < 8$; its 8 faces with antipodal-free vertex sets are the 4 vertices and the 4 edges), and the octahedron ($k = 3$: $6 < 16$; all 26 antipodal-free vertex sets are faces: 6 vertices, 12 edges, 8 triangles). For $k = 0$ the only polytope is a point, with $1 \neq 2^1$ vertex.

Remark 3 (Other readings). 1. *Center.* The cross-polytope is symmetric about the origin, so it is centrally symmetric about some center under any convention.

2. *k as the neighborliness parameter.* If “ k -dimensional” is read as “cs- k -neighborly” in arbitrary dimension, then C_k^Δ is a cs- k -neighborly polytope with $2k < 2^{k+1}$ vertices.
3. *Lattice vs. real polytopes.* Our example has integer vertices. A counterexample in a subclass is a counterexample to the universally quantified claim.
4. *Facets instead of vertices.* This is not what is written, but it also fails. C_k^Δ has $2^k < 2^{k+1}$ facets. Under the literal definition, the cube $[-1, 1]^k$ qualifies and has $2k$ facets.

5 The Chinese reading: dimension $2k$

The Chinese text reads: the minimal number of vertices of a $2k$ -dimensional cs-neighborly polytope is 2^{k+1} . As a universal statement, this is

$$\forall k \geq 1 : \min\{f_0(P) : P \text{ a } 2k\text{-dimensional cs-neighborly polytope}\} = 2^{k+1}, \quad (2)$$

and in particular it contains the universal lower bound: every $2k$ -dimensional cs-neighborly polytope has at least 2^{k+1} vertices.

Theorem 4. *Claim (2) holds for $k = 1, 2$ and fails for every $k \geq 3$, under each reading of “cs-neighborly”. The smallest counterexample is the 6-dimensional cross-polytope C_6^Δ , which has property (F) and $12 < 16 = 2^4$ vertices.*

Proof. By Proposition 5 (below), the minimum in dimension $2k$ is $4k$, under every reading. We have $4k = 2^{k+1}$ for $k = 1, 2$ ($4 = 4$, $8 = 8$), and $4k < 2^{k+1}$ for $k \geq 3$ ($12 < 16$, and if $4k < 2^{k+1}$ then $4(k+1) = 4k + 4 < 2^{k+1} + 2^{k+1} = 2^{k+2}$, using $4 \leq 2^{k+1}$). For $k \geq 3$, C_{2k}^Δ is a counterexample (Theorem 1). $\square$

So the Chinese version is also false as a statement about all k , but its first two instances are true; a counterexample needs dimension at least 6. Lean proves `conjecture_00000001564_false_2k`: for every notion N that C_6^Δ satisfies, $\neg \forall k \geq 1, \text{MinVerticesIs } N(2*k)(2^{k+1})$, with instances for the literal definition, property (F), cs- j -neighborliness for every j , and cs-3-neighborliness (Grünbaum’s cs- $\lfloor d/2 \rfloor$ -neighborliness for $d = 6$). The case $k = 3$ needs only the lower-bound half, which C_6^Δ violates. The truth of the instances $k = 1, 2$ rests on Proposition 5, which is not formalized in dimensions 2 and 4.

6 The true minimum and the Hanner clause

Proposition 5. *Every k -dimensional centrally symmetric polytope has at least $2k$ vertices. Equality holds exactly for the affine images of C_k^Δ , and these have property (F). Hence, under each reading above, the minimal number of vertices of a k -dimensional cs-neighborly polytope is $2k$, attained exactly by the affine cross-polytopes.*

Proof. Translate the center to 0, so $V = -V$. Since $0 = \frac{1}{2}(v + (-v))$ lies in the affine hull, the affine hull equals the linear span of V , which is therefore $\mathbb{R}^k$. Choose a basis $v_1, \dots, v_k \in V$. The $2k$ points $\pm v_i$ lie in V and are distinct: $v_i \neq 0$, and $v_i = \pm v_j$ for $i \neq j$ contradicts independence. If $|V| = 2k$, then $V = \{\pm v_i\} = AV_k$ with A the invertible linear map $e_i \mapsto v_i$. An invertible affine map preserves faces, central symmetry and antipodality, so AC_k^Δ has property (F). Conversely, every affine image of C_k^Δ has $2k$ vertices. $\square$

Remark 6 (A “maximal” reading). If “minimal” were a slip for “maximal”, the claim would still fail for the k -dimensional reading. Under the literal definition there is no maximum (centrally symmetric $2m$ -gons exist for every m). Under Grünbaum’s cs-2-neighborliness, Linial and Novik [3] proved that a cs-2-neighborly d -polytope has at most 2^d vertices, so 2^{k+1} is not attained. This remark is not formalized.

Hanner polytopes. Hanner polytopes are generated from the segment $[-1, 1]$ by Cartesian products $P \times Q$ and free sums $P \oplus Q = \text{conv}(P \times \{0\} \cup \{0\} \times Q)$. For centrally symmetric P, Q with 0 in the interior, $f_0(P \times Q) = f_0(P)f_0(Q)$ and $f_0(P \oplus Q) = f_0(P) + f_0(Q)$, and dimensions add. By induction, a d -dimensional Hanner polytope satisfies $2d \leq f_0 \leq 2^d$. The base case is the segment, $f_0 = 2$. For the product step, $d_1 + d_2 \leq 2d_1d_2$ and $2^{d_1}2^{d_2} = 2^{d_1+d_2}$. For the free-sum step, $x + y \leq xy$ for $x, y \geq 2$ gives $2^{d_1} + 2^{d_2} \leq 2^{d_1+d_2}$. So no Hanner polytope has 2^{d+1} vertices, and the conjecture’s two clauses cannot both hold if “extremal bodies” means the bodies attaining the minimum. The cube (2^d vertices) and the cross-polytope ($2d$ vertices) are Hanner. By Proposition 5, the true minimizers are affine cross-polytopes, which are Hanner up to linear equivalence. So the corrected statement “minimum $2k$, attained by (affine) Hanner polytopes” is true. (The conjecture may be a garbled form of Kalai’s 3^d conjecture on the total number of faces of cs polytopes, where Hanner polytopes are the conjectured extremal bodies. That is a different statement.)

7 Lean formalization

The project `lean4/` (Lean 4.19.0, core library only, no Mathlib) defines all objects from scratch.

- Points of $\mathbb{Z}^k$ are lists of k integers, with the scalar product `dot`. A polytope is given by its vertex list `V`.
 - `IsVertex V v`: v is the unique maximizer over V of some functional.
 - `ConvexPosition V`: no repetitions, and every point is a vertex, so the polytope has `V.length` vertices.
 - `IsFace V S`: S is exactly the set of maximizers of some functional.
 - `CentSymm V`: $V = -V$.
- `FullDim k V` is a dimension certificate: points $a, b_1, \dots, b_k \in V$ and functionals f_i with $f_i(b_j - a) \neq 0 \iff i = j$. `fullDim_linIndep` and `fullDim_sound` prove that the $b_j - a$ are then linearly independent (the same argument works over $\mathbb{R}$). Together with all points lying in $\mathbb{Z}^k$, this gives `IsPolytopeOfDim k V`.
- The three readings are `StatementCsNeighborly` (the literal definition), `CsNeighborlyUpTo j` (Grünbaum’s cs- j -neighborliness) and `CsNeighborlyFull` (property (F)). `full_upTo` shows that (F) implies the second for every j , and `statement_def_of_cs` shows that every nonempty cs polytope satisfies the first.
- `MinVerticesIs N k n` says that the minimum is attained and is a lower bound; `MinClause N` is the clause $\forall k \geq 1, \text{MinVerticesIs } N \text{ } k \text{ } (2^{k+1})$.
- `crossFacts_1` to `crossFacts_4` and `crossFacts_6` check, by `decide` for the generated list `cross k`:
 - length $2k$, no repetitions, central symmetry;

- the vertex property (witness $c = v$) and the dimension certificate;
- for every subset S (every sublist), that if S is nonempty and antipodal-free, then the maximizers of $\sum S$ are exactly S .

`full_of_sublists` transfers the last check from sublists to arbitrary subsets.

- Main theorems:

- `conjecture_00000001564_false` : $\neg \text{MinClause StatementCsNeighborly}$;
- `conjecture_00000001564_false_grunbaum` (j), for every j ;
- `conjecture_00000001564_false_full`;
- `conjecture_00000001564_false_any`, for any notion satisfied by the octahedron;
- `conjecture_00000001564_false_param`, the reading in Remark 3(2);
- `conjecture_00000001564_false_2k`: the Chinese $2k$ -dimensional reading (Section 5), via `crossFacts_6`, with four instances (suffixes `statement`, `full`, `grunbaum`, `cs3`);
- `fails_each_k`: failure at each $k = 1, 2, 3, 4$;
- `conjecture_00000001564_conjunction_false`: the conjunction with any Hanner clause is false.

- Non-vacuity:

- `true_min_k1` proves a true instance, `MinVerticesIs` $\leq 1/2$, including its lower bound.
- A centrally symmetric hexagon is a 2-polytope satisfying the literal definition but not cs-2-neighborliness (`hexagon_not_csNeighborly`).
- A set with an interior point is not in convex position.

- `HannerCount` records the (dimension, vertex number) pairs generated by segments, products and free sums. `hanner_bounds` proves $2d \leq v \leq 2^d$, and `no_hanner_with_2pow_succ` proves $v \neq 2^{d+1}$.

There is no `sorry`, `native_decide` or added axiom. `#print axioms` reports only `propext`, `Quot.sound` and (for the generic linear-independence lemma) `Classical.choice`.

Scope.

- Lean checks the cross-polytope for $k = 1, 2, 3, 4, 6$; one k suffices to refute the clause. The case of general k is Theorems 1 and 2 above, and `verify.py` checks $k \leq 6$.
- The general lower bound $2k$ of Proposition 5 is proved here. It is formalized only for $k = 1$, and `verify.py` checks it for lattice sets in $\{-1, 0, 1\}^k$, $k = 2, 3$.
- For Hanner polytopes, only the vertex-count recursion is formalized. The facts $f_0(P \times Q) = f_0(P)f_0(Q)$ and $f_0(P \oplus Q) = f_0(P) + f_0(Q)$ are standard, and `verify.py` confirms them on explicit point sets.

The integer functionals used as witnesses are real functionals, so every Lean predicate implies its real counterpart. In particular, `conv(cross 3)` is a genuine 3-dimensional cs polytope in $\mathbb{R}^3$ with 6 vertices and property (F).

8 Independent check (verify.py)

The Python script uses only the standard library and a method different from the Lean proof.

- For $k \leq 6$, it enumerates all functionals $c \in \{-1, 0, 1\}^k \setminus \{0\}$, and for $k \leq 3$ also all $c \in \{-2, \dots, 2\}^k \setminus \{0\}$. It collects every maximizer set and confirms that this family is exactly the family of antipodal-free vertex subsets (2, 8, 26, 80, 242, 728 for $k = 1, \dots, 6$). It also computes the dimension as an exact rank.
- It checks every centrally symmetric point set in $\{-1, 0, 1\}^k$ ($k = 2, 3$) that spans $\mathbb{R}^k$: each gives a polytope with at least $2k$ vertices.
- It builds all 1, 2, 6, 22, 90, 394 explicit Hanner polytopes of dimension $1, \dots, 6$ and confirms that their vertex numbers lie in $[2d, 2^d]$ and never equal 2^{d+1} .

9 Reproduction

```
cd lean4 && lake build          # Lean 4.19.0, core only
cd .. && python3 verify.py      # Python 3 standard library
pdflatex report.tex && pdflatex report.tex
```

References

- [1] B. Grünbaum, *Convex Polytopes*, 2nd ed., Springer GTM 221, 2003 (chapter on centrally symmetric polytopes).
- [2] O. Hanner, Intersections of translates of convex bodies, *Math. Scand.* 4 (1956), 65–87.
- [3] N. Linial and I. Novik, How neighborly can a centrally symmetric polytope be?, *Discrete Comput. Geom.* 36 (2006), 273–281.
- [4] G. Kalai, The number of faces of centrally-symmetric polytopes, *Graphs Combin.* 5 (1989), 389–391.